

CORE_STATE — Stable Core Snapshot

Status

State: STABLE **Phase:** Core finalized, simulation-validated **Date:** 2026-02-01

This document fixes the exact working state of the engine core. From here, all further work (visuals, screenshots, real chart input, Telegram bot integration) must **build on top**, not rewrite core logic.

Engine Purpose

The core engine decides **whether trading is allowed or blocked** based on internal state, learning phase, authority, expectancy, and cooldown logic.

It does **NOT**: - Generate BUY/SELL entries - Analyze real charts - Depend on screenshots or indicators

It **DOES**: - Control permission to trade - Learn confidence/authority over time - Enforce cooldown **only after real trades**

Core Concepts

Modes

- **BOOTSTRAP** — learning sandbox, no real risk
- **LIVE** — production logic

Decisions

- LEARN_BOOTSTRAP
- ALLOW_TRADE
- BLOCK_COOLDOWN
- NO_TRADE

Flags

- learning: True | False
 - execute: True | False
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Cooldown Rule (FIXED)

! Cooldown is applied **ONLY** after `ALLOW_TRADE` with `execute=True`.

Never on startup. Never globally. Never if no trade was executed.

This bug is resolved and verified by simulation.

Simulation Results (Reference Run)

Total steps: 20

Metric	Value
Wins	3
Losses	3
Final Authority	0.15
Expectancy	0
Cooldown behavior	✅ Correct

Observed pattern: - Bootstrap trades allowed only in sandbox - Transition to LIVE after first win - Cooldown triggers strictly after executed trades - Authority increases on wins, decreases on losses

Invariants (DO NOT BREAK)

- Engine must run with **empty or missing JSON state**
 - Simulation must not depend on filesystem persistence
 - `decision != final_signal`
 - Cooldown never blocks the first valid trade
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Files Considered CORE

These files form the immutable core:

- `decision_engine.py`
- `engine_state.py`
- `simulation_runner.py`

Any changes to these require: 1. Explicit version bump 2. New simulation proof 3. New CORE_STATE snapshot

What Comes Next (Allowed Work)

✅ Screenshot ingestion ✅ Chart vision module ✅ Signal generator (BUY/SELL) ✅ Telegram bot ✅ Persistence & memory

❌ Core rewrite ❌ Logic simplification ❌ Cooldown redesign

Rollback Instruction

If anything breaks: 1. Restore core files 2. Delete all `.json` state 3. Run `simulation_runner.py` 4. Validate against this document

This is the anchor point. No more core chaos. Only forward motion. 